Design and Implementation of a Low-cost Carbon Fiber Bionic Arm with Multi-modal Sensor Fusion and IoT-Enabled Adaptive Control

Ahmed Bahaa El-Din, Saad Elsayed Saad

***Abstract*—Contemporary prosthetic arms are often constrained by non-intuitive control mechanisms, excessive weight, and high cost. This paper presents the design and implementation of a low-cost, biomechatronic prosthetic arm utilizing a lightweight carbon fiber frame to achieve a 40% weight reduction (350g) compared to traditional 3D-printed frameworks. To overcome the high noise floor inherent in custom-built surface electromyography (EMG) sensors, a hybrid multimodal sensor fusion framework is introduced. The system implements a deterministic gating logic where a digital moving average filtered EMG signal acts strictly as an asynchronous activation trigger, while an armband pressure matrix tracks muscle expansion to determine specific grasp intents. An ESP32 microcontroller processes the sensory data locally, driving seven independent micro-actuators via an adaptive proportional-integral-derivative (PID) control loop to ensure smooth, natural object manipulation. An IoT framework enables remote monitoring and clinician-led data logging. The prototype successfully delivers responsive, low-latency actuation under 500ms at an approximate cost of $310 USD.**

***Keywords*— Myoelectric control, Prosthetic manipulation, Multimodal sensing, Embedded rehabilitation systems, Human–machine interface, Assistive robotics.**

I. INTRODUCTION

THE Upper limb loss carries profound consequences for independence, since the hand underpins nearly every activity of daily living, and traumatic amputation remains its leading driver. Conventional body-powered and cosmetic prostheses often fail to restore functional dexterity, intuitive control, or meaningful feedback, with studies reporting abandonment rates of 26% for body-powered and 23% for electric devices [4]. While surface electromyography (sEMG) combined with pattern recognition has matured, persistent gaps—such as unintuitive control mappings and fragility under real-world variability—continue to keep most of these advances confined to the laboratory [1], [2], [3].

Beyond these technical hurdles, the global need for accessible assistive technology is staggering. With over 2.5 billion people requiring at least one assistive product, the World Health Organization and UNICEF (2022) report that access remains severely limited in low- and middle-income countries, with some regions seeing access rates as low as 3% [5]. This trend is mirrored locally in Egypt, where data from the Central Agency for Public Mobilization and Statistics (CAPMAS) indicates a disability prevalence of approximately 11% [6]. These figures underscore a critical socioeconomic gap in the availability of affordable, high-quality, and accessible biomedical solutions.

Addressing these shortfalls requires a convergent design that fuses heterogeneous biosignal and environmental sensing, lightens the mechanical load through composite materials such as carbon-fiber-reinforced laminates, and embeds the device inside a wider IoT service ecosystem for monitoring, recalibration, and personalization [7], [8], [9]. The mechanical architecture of the proposed hand builds upon the open-source compliant prosthetic design of Akhtar et al. [10], originally fabricated from PLA and silicone, extending it through carbon fiber composite construction, wrist rotation via a stepper motor, a six-axis sensor fusion framework (EMG, IMU, IR, FSR, Velostat armband, and temperature), empirical EMG threshold calibration, and an IoT framework comprising a mobile application and clinical web portal. This paper presents a multisensory-controlled bionic upper-limb prosthesis that integrates an accelerometer, infrared sensor, electromyography channel with an AI-driven feature-extraction and classification layer, distributed force-sensitive resistors (including an armband coupled to the EMG electrodes), and temperature sensing within a carbon-fiber, stainless-steel, and polymer hybrid chassis actuated by six DC metal-geared motors and one mini stepper motor for wrist articulation. Intelligence here is confined deliberately to EMG interpretation rather than end-to-end autonomous decision making, while a dedicated Internet-of-Things layer—accessible through a mobile application and an accompanying web platform—provides remote monitoring, on-the-fly tuning, news dissemination, and a customization channel for users to commission prostheses fitted to their individual clinical, occupational, and aesthetic needs. By unifying multimodal sensing, robust composite construction, and connected service infrastructure within one platform, the proposed system targets the persistent gap between laboratory-grade myoelectric research and the everyday usability demanded by amputees [9].

(*Corresponding author: Ahmed Bahaa El-Din*).

Ahmed Bahaa El-Din is with the Department of Electronics and Communication Engineering, Higher Institute of Engineering and Technology Al-Obour , PO Box 27, Al Obour City Egypt (e-mail: 221399@oi.edu.eg).

Saad Elsayed Saad is with the Department of Electronics and Communication Engineering, Higher Institute of Engineering and Technology Al-Obour, PO Box 27. He is now with the Department of Engineering and Technology, Obour University For Science & Technology, Al-Obour, (e-mail: saad.elsayed@oust.edu.eg).

II. MATERIALS AND METHODS

A. Mechanical Structure and Material Fabrication

The forearm shell, palm plate, and dorsal hand enclosure of the AB-Limb were custom-fabricated from a carbon-fiber-reinforced epoxy laminate, as shown in Fig. 1. Fabrication was executed via Vacuum-Assisted Resin Transfer Molding (VARTM), a process in which a flexible vacuum bag seals a single-sided rigid mold so that atmospheric pressure drives resin through a dry carbon-fiber preform, allowing tight control of the fiber-to-resin ratio and evacuation of internal micro-voids without requiring autoclave pressure [11].

Fig. 1. Final AB-Limb prototype showing the assembled carbon fiber bionic arm alongside raw carbon fiber composite sheets used in fabrication.

The hand mold geometry was adapted from the open-source compliant prosthetic design of Akhtar et al. [10], which was originally developed for silicone casting. In this work, the same mold architecture was repurposed for VARTM carbon fiber lamination: woven carbon-fiber plies were dry-laid over each mold, sealed under vacuum bagging film, and infused with a room-temperature-curing epoxy system. The modular component geometries — including wrist shell, palm plate, finger base blocks, and forearm cylinder — were further refined using parametric OpenSCAD scripts to generate gear profiles and mold optimizations prior to casting, as shown in Fig. 2 and Fig. 3.

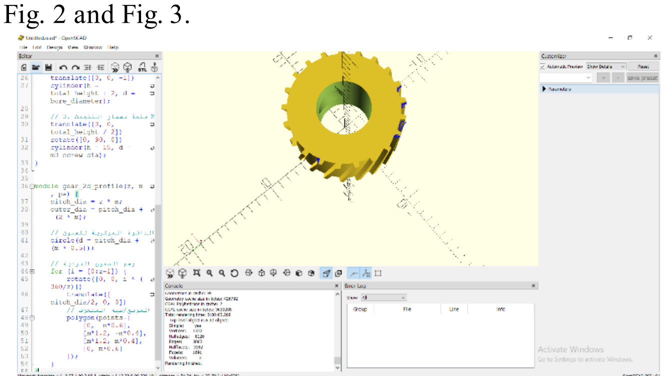

Fig. 2. Parametric computer-aided design and modular structural gears optimization using OpenSCAD environment.

The resulting composite structure achieved a tensile strength of approximately 600 MPa at a total structural mass of 350 g, representing a 40% weight reduction compared to equivalent aluminum or standard 3D-printed PLA frameworks [12], [13], [14]. This lightweight architecture reduces inertial resistance during movement, enhancing user mobility and reducing fatigue at the residual limb interface. A 1-Degree-of-Freedom wrist rotation unit driven by a miniature stepper motor was integrated along the forearm axis to provide pronation–supination capability absent from earlier single-anchor prosthetic frames [10].

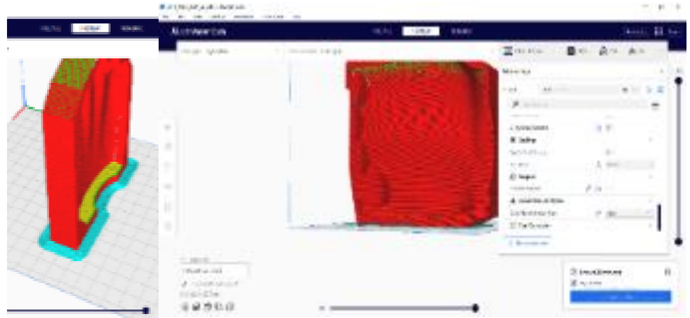

Fig. 3. Physical hand mold adapted from the open-source design of Akhtar et al. [10], originally developed for silicone casting and modified in this work for VARTM carbon fiber lamination.

B. Actuators and Hybrid Drive Integration

Initial prototyping evaluated a direct-drive configuration in which six N20 DC micro-motors were driven through three L293D H-bridge chips cascaded via two cascaded 74HC595 8-bit shift registers were controlled by arduino uno [19],To manage these six channels while minimizing arduino GPIO consumption, as shown in Fig. 4. This architecture reduces the entire six-motor control bus of 4 pins for each motor to a three-wire serial interface — data, clock, and latch — from the microcontroller, electrically isolating the Arduino digital logic rail from inductive kickback during motor stalls .

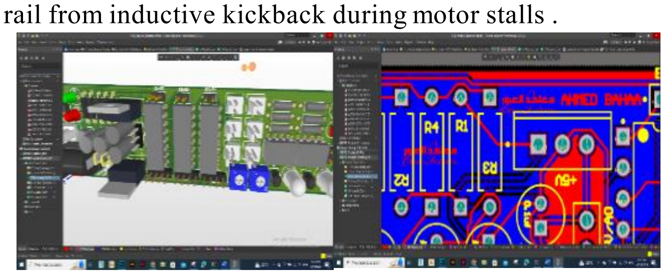

Fig. 4. Schematic diagram of the intermediate shift register topology interfaced with localized L293D motor drivers.

While this arrangement confirmed mechanical feasibility, positional accuracy was insufficient for reliable grasp control and the GPIO demand — even after shift-register demultiplexing — constrained the available pins for the remaining sensor subsystems. These limitations motivated a redesign centered on the ESP32 and a hybrid actuation topology described below.

In the refined system, independent flexion and extension of the five fingers is achieved through six N20-series brushed DC micro-motors with high-ratio metal gearboxes.

Two motors actuate the thumb across two degrees of freedom, while the remaining four drive the index, middle, ring, and little fingers through linkages and custom epoxy-cast gears, as shown in Fig. 2 and Fig. 1.

A seventh channel drives the 1-DOF wrist rotation module via a 5V miniature stepper motor controlled through a ULN2003A driver board [15], [16].

Standard N20 units ship without integral closed-loop position feedback, which typically requires adding bulky optical or magnetic encoders to achieve controlled directional behavior

[16], [17]. To resolve this without inflating the device form factor, the internal control board of an SG90 micro-servo — including its feedback potentiometer and PWM conditioning circuit — was decoupled and integrated directly onto the N20 motor chassis, as illustrated in Fig. 5. The potentiometer is monitored continuously by the ESP32 PWM; when the wiper crosses a predefined directional threshold, a stop pulse is issued and the motor reverses, enabling reliable bidirectional control over each channel independently. This modification eliminates secondary encoders while reducing the per-motor control requirement to a single PWM line, allowing all six fingers to be addressed through only six control signals total.

Fig. 5. Electro-mechanical modification layout showing the integration of decoupled SG90 servo control circuitry onto the N20 chassis.

the PWM commands are generated as hardware-timer pulses on the ESP32 for non-blocking real-time control, and are dynamically scaled by incoming force sensor and IR proximity readings through a local PID loop to suppress oscillation and overshoot during object approach [18].

C. Multi-Modal Sensor Integration and Signal Conditioning

1) Custom Surface Electromyography (EMG) System:

To maintain the cost-effective architecture of the AB-Limb, a custom single-channel active surface EMG (sEMG) acquisition front-end was developed (Fig. 6a), adapting the open-source *OpenEMG* hardware architecture developed by CharlesLabs as a structural baseline. The physical acquisition layout utilizes two Ag/AgCl surface electrodes with a 2 cm inter-electrode pitch positioned over the belly of the target residual-limb muscle group. A third reference electrode is placed on a nearby bony prominence to suppress common-mode environmental noise.

The differential biopotential is initially fed into a three-op-amp instrumentation amplifier front-end built around an LM324 quad operational amplifier (Fig. 6b). This configuration provides a high input impedance (>10 MΩ) to prevent skin-electrode loading artifacts while maximizing the Common-Mode Rejection Ratio (CMRR) to suppress powerline pickup and motion-induced baseline drift.

To enhance signal conditioning beyond the baseline open-source design, a secondary custom active filtering stage was integrated immediately following the instrumentation amplifier. This stage, engineered utilizing an LM358 dual operational amplifier, implements an active bandpass filter restricting the passband strictly between 20 Hz and 450 Hz. This optimization isolates the canonical sEMG spectral envelope while effectively rejecting low-frequency motion artifacts and high-frequency electronic noise.

The conditioned analog signal is then sampled by the central microcontroller. In firmware, a digital notch filter (50/60 Hz) is executed as an additional guard against grid interference. Following full-wave rectification, the signal is processed via a digital moving-average filter—implemented as a sliding window accumulator over N consecutive ADC samples—to yield a stable, low-latency amplitude envelope suitable as an asynchronous deterministic trigger [1], [2], [3].

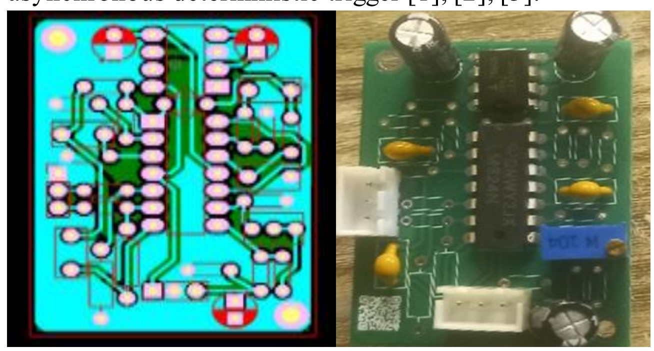

Fig. 6a. Custom-built active surface EMG sensor PCB layout.

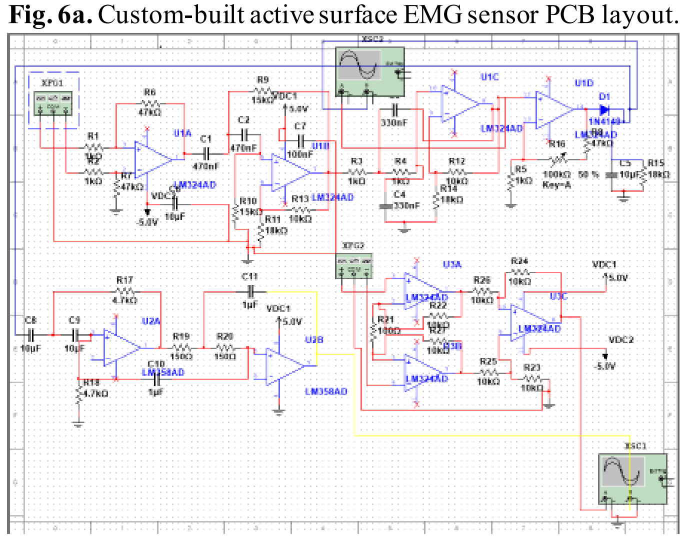

Fig. 6b. Custom-built active surface EMG sensor schematic hardware showing the LM324 instrumentation amplifier front-end and LM358 bandpass filtering stage.

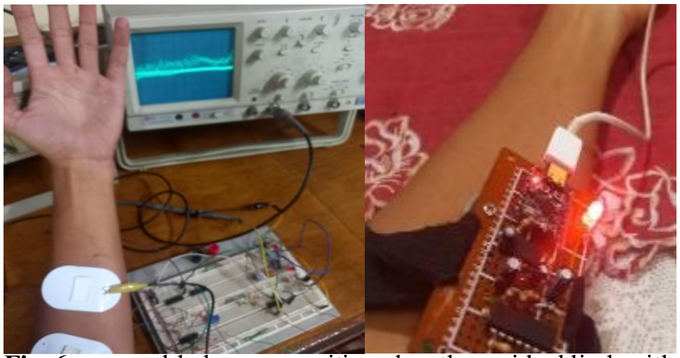

Fig. 6c. assembled sensor positioned on the residual limb with electrode connections,complete EMG acquisition unit integrated with ESP32-C3 controller for test.

Initial EMG data acquisition sessions revealed three critical calibration challenges that were systematically mitigated. First, a data-labeling anomaly occurred where a spurious third class label ('testing') was inadvertently introduced during token collection, degrading classifier performance to baseline chance level (~50%). This was resolved by filtering the training dataset to enforce strict binary classification (Contracted vs. Relaxed). Second, a structural class imbalance between contraction and relaxation samples was corrected using data resampling techniques to eliminate model bias toward the majority class. Third, an initial digital signal processing (DSP) misconfiguration utilized an improper high-pass filter setting that introduced phase distortion and attenuated critical low-frequency contraction envelopes while passing extraneous high-frequency electronic noise (Fig. 7). This was corrected by optimizing the digital filter coefficients to stabilize the baseline envelope and ensure robust intent recognition.

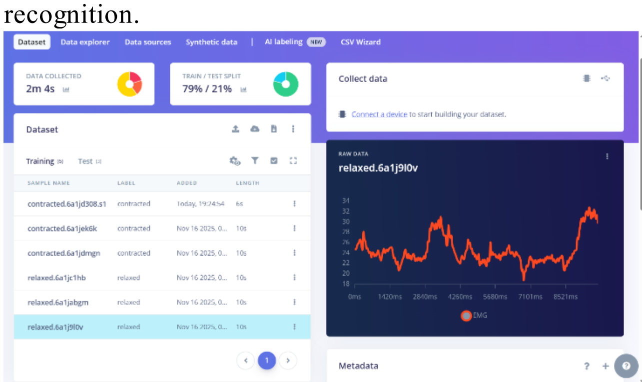

Fig. 7. Initial DSP configuration showing high-pass filter at 19 Hz cut-off — this setting passed high-frequency noise and produced an unstable filtered output, motivating the filter redesign.

These issues were resolved sequentially: the 'testing' samples were deleted from the dataset, recording sessions were repeated to achieve balanced class durations, and the filter was reconfigured to a low-pass type with a 50 Hz cut-off frequency and FFT length of 512, with StandardScaler normalization enabled. Following these corrections, model training over 30 epochs with a learning rate of 0.0005 achieved 100% classification accuracy with a final loss of 0.00 on both training and validation sets, as confirmed by TensorBoard convergence curves Fig.8. Fig. 9. The INT8-quantized model deployed on the ESP32 demonstrated an inference latency of 1 ms, a peak RAM consumption of 2.4 KB, and a Flash footprint of 35.8 KB.

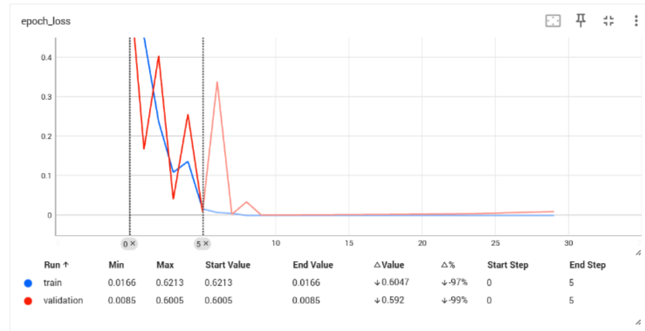

Fig. 8. TensorBoard epoch loss curves, prior to data cleansing, showing validation loss divergence and 50% accuracy indicative of overfitting.

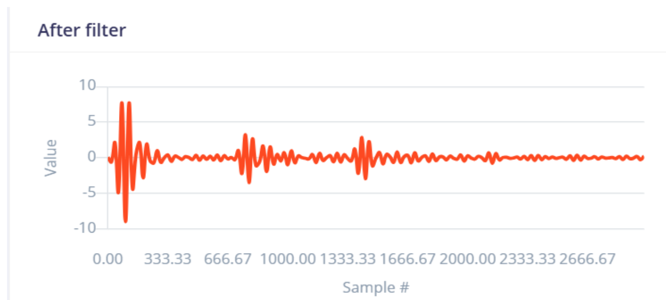

Fig. 9. following data balancing and DSP correction, showing co-convergence of training and validation loss to 0.00 within 10 epochs, confirming successful generalization.

Subject-specific activation thresholds were determined empirically using the Edge Impulse Studio platform as a signal visualization tool. Raw sEMG signals were captured under two controlled conditions — full voluntary muscle contraction and complete rest — and the resulting amplitude envelopes were inspected to extract the peak activation level and the quiescent baseline, as illustrated in Fig. 10. The activation threshold was iteratively refined through empirical calibration sessions: the subject performed repeated contraction and rest cycles while the threshold constant in firmware was manually adjusted until reliable, false-positive-free triggering was achieved for that individual's signal profile. These empirically determined bounds were encoded as fixed threshold constants in the ESP32 firmware, providing a patient-calibrated binary trigger that avoids the latency and computational overhead of real-time multi-class classifiers.

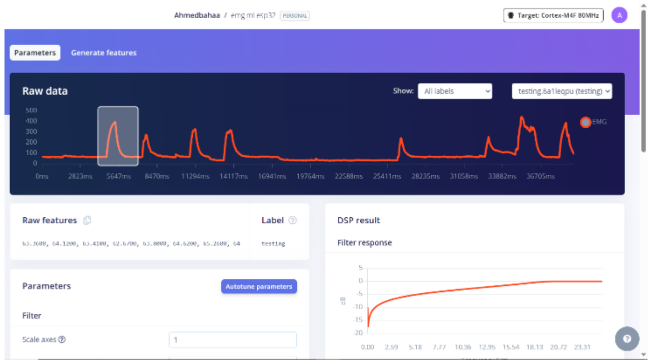

Fig. 10. Raw sEMG signal acquired via Edge Impulse Studio showing repeated voluntary contraction events (amplitude spikes reaching 0–450 ADC units) interspersed with quiescent rest periods, used to empirically determine the subject-specific activation threshold encoded in ESP32 firmware.

To suppress false-positive triggers arising from EMG drift or motion artifact, actuator activation requires simultaneous confirmation from the 2×2 Velostat armband matrix Fig. 11.

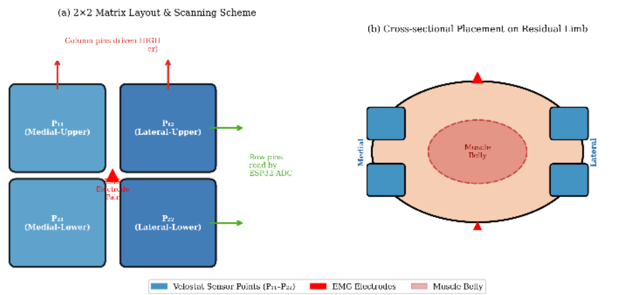

Fig. 11. Armband pressure matrix showing sensor placement medial and lateral to the primary EMG electrode pair over the muscle belly, and the pulse-scanning readout scheme implemented.

Fig. 12. layout of the armband and EMG electrode pair over the muscle belly.

the motor command is only issued when both the EMG threshold and at least one armband pressure point are crossed within the same sampling window, implementing a dual-verification logic gate in firmware [22], [23].The four armband sensing points are positioned at two anatomical landmarks: medial and lateral to the primary EMG electrode pair, directly over the muscle belly Fig. 12., so that voluntary contraction produces a consistent bilateral pressure signature.

2) Spatial Armband Pressure Matrix and Custom Force Sensors:

To resolve the false-positive triggering that arises from raw sEMG drift, the AB-Limb integrates a tactile sensing layer with two distinct physical roles: an internal intent validator and an external contact verifier [24]. The internal layer is a custom 2×2 armband pressure matrix fabricated from flexible Velostat sheet, a carbon-loaded polymeric foil whose resistance decreases under pressure, making it a low-cost piezoresistive transducer for wearable sensing [22], [25].

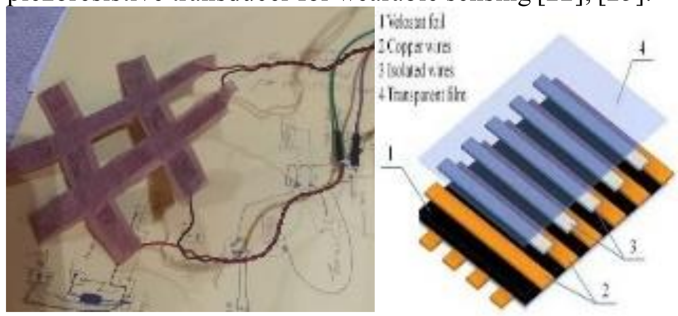

Fig. 13. Structural layout of the custom Velostat-based spatial armband pressure matrix.

Each column of the matrix was sequentially driven HIGH while the orthogonal rows were read by the ESP32 ADC, implementing a fast pulse-scanning readout that suppresses ghosting during simultaneous multi-point contact. The raw readings were smoothed with a first-order digital low-pass filter with $\alpha = 0.08$ to yield a stable mechanical image of muscle bulging under voluntary contraction, providing an intent channel independent of the electrical noise floor that limits surface EMG [24], [25], [26].

For external contact verification, four custom force-sensing resistors measuring 6 mm × 6 mm were embedded into the distal phalanges of the prosthetic fingertips Fig. 14. each FSR was wired into a 10 kΩ pull-down voltage divider so that mechanical load maps linearly to an analog voltage readable by the ESP32 ADC [27], [28]. The force reading was passed through a Force-to-Velocity algorithm that progressively reduces drive PWM as grip force increases, preventing brittle objects from being crushed during pinch closure. A localized PWM-driven vibration motor against the user's skin was triggered on contact confirmation to close the sensory loop [29], [30].

Fig. 14. Miniaturized 6mm x 6mm custom fingertip force sensor integration and voltage divider conditioning circuit.

3) Kinematic Motion Processing Unit (MPU):
An MPU6050 six-axis inertial measurement unit, combining a tri-axis accelerometer and a tri-axis gyroscope on a single chip, was embedded inside the dorsal hand enclosure to provide orientation-aware control that operates independently of, and complementary to, the sEMG channel [29], [31]. Raw angular data from the gyroscope was fused with accelerometer-derived gravity references through a complementary filter to obtain a stable tilt estimate along the wrist's principal bending axis Fig. 15. When gesture-control mode is selected through the mobile application, the residual-limb tilt is fed through the same PID regulator used for motor commutation: forward tilt drives uniform finger flexion, backward tilt drives uniform extension, and a dead-band around the level orientation locks the actuators in a steady-state hold [32], [33], [34].

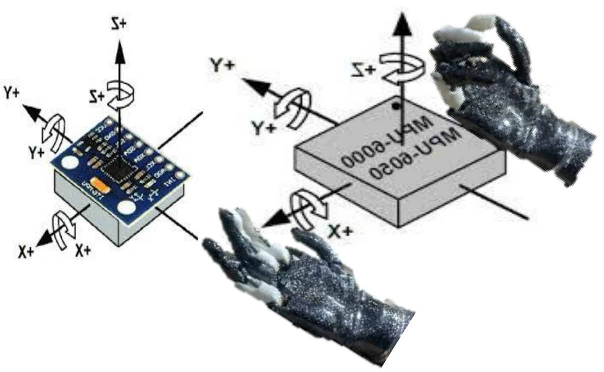

Fig. 15. MPU6050 orientation tracking axes and the corresponding multi-motor coordinated control mapping.

4) Infrared (IR) Proximity and Collision Avoidance Gating:
A short-range infrared emitter-photodiode pair mounted on the palmar surface of the prosthesis continuously samples the distance between the terminal device and any approaching object. The photodiode output is conditioned by an LM358-based Fig. 16. comparator that yields a binary trigger as the object enters a pre-set safety envelope, and the smoothed analog channel is fed into the same PID loop that governs motor velocity so that approach speed is scaled inversely with distance, decelerating the actuators near object boundaries [29], [35], [27], [36]. This channel also acts as a hardware-interrupt-driven reflexive layer: a rapid proximity transition during a hold-and-release state bypasses the main sEMG/IMU threads and fires an emergency close command that mimics a biological protective reflex [30], [37].

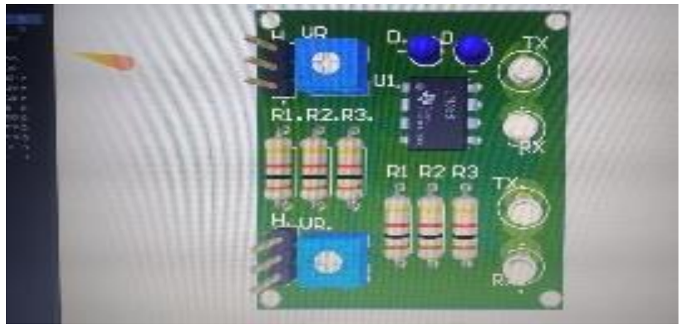

Fig. 16. PCB layout of the LM358-driven infrared proximity sensing circuit.

5) Thermal Monitoring Layer:
An NTC thermistor with β = 3950 was embedded at the residual-limb interface inside the socket wall, selected over non-contact infrared modules such as the MLX90614 because its small form factor and direct skin contact provide higher accuracy in the clinically relevant 35°C to 42°C range [38], [39], [40]. The thermistor was paired with a 10 kΩ balance resistor to form a voltage divider readable by the ESP32 ADC, and the nonlinear resistance-to-temperature relationship was linearized in firmware using the Steinhart–Hart equation:

$$\frac{1}{T} = A + B\ln(R) + C\,(\ln(R))^3 \qquad (1)$$

where T is the absolute temperature in kelvin, R is the computed thermistor resistance, and A, B, and C are coefficients derived from the manufacturer datasheet (β = 3950). Because the ESP32's successive-approximation ADC exhibits nonlinearity near the 0 V and 3.3 V saturation margins, an oversampling and averaging routine of 10–20 samples per reading was executed to suppress transient electronic noise and stabilize the calibrated output [41], [42]. Thermal readings are streamed to the cloud IoT portal and trigger an alert if the residual-limb skin temperature exceeds the physiologically safe envelope, guarding against the in-socket heat accumulation linked to skin irritation and prosthesis abandonment [43], [44].

D. Control System Architecture, IoT Infrastructure, and Power Management

1) Central Controller and System Architecture:
The core processing hub of the AB-Limb is an ESP32 DevKit microcontroller, chosen for its dual-core 240 MHz architecture, native Wi-Fi and Bluetooth Low Energy radios, and its demonstrated capacity to host real-time sEMG and sensor-fusion pipelines on a low-cost embedded platform [21], [45]. The firmware is organized as a state-triggered actuation engine: grip selection, sensor gating, and actuation windows are exposed as runtime-tunable parameters that the user can adjust through the mobile companion application rather than through low-level firmware reflash, allowing patients to adapt the prosthesis to changing daily-life activities without any code modification [46], [21].

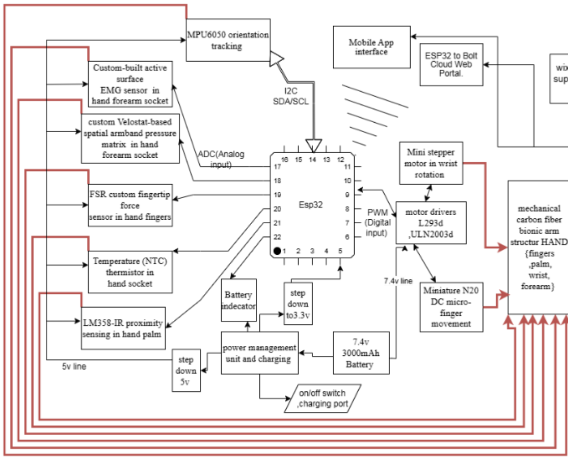

Fig. 16. System block diagram showing the ESP32 core architecture, multi-sensor interfacing layers, and peripheral distribution.

2) Cloud IoT Telemetry and Remote Clinical Portal:

Filtered sEMG envelopes, force-matrix readings, IMU orientation logs, and thermal samples are encrypted and transmitted over Wi-Fi to a centralized cloud portal built around the Bolt IoT framework, exposing WebSocket, MQTT, and HTTP endpoints that allow clinicians to asynchronously review biomechanical signal histories, track muscle-health progression, and adjust validation thresholds remotely [9], [46], [47], [48]. The same portal hosts a public news and updates channel and a customization intake form through which users can commission prostheses fitted to their individual clinical, occupational, and aesthetic requirements, closing the Industry 4.0 loop between physical actuation and digital service infrastructure [49].**Fig. 17**.

Fig. 17. Cloud IoT web portal interface showing real-time EMG signal monitoring dashboard, remote clinical threshold adjustment panel, and patient customization intake form.

3) Power Management System:

Mobile operation is sustained by a lightweight 7.4 V two-cell lithium-polymer battery pack with a high discharge C-rating sufficient to absorb the transient inrush currents drawn during simultaneous multi-motor grasping routines. A dual-stage synchronous buck DC-to-DC converter topology supplies a regulated 3.3 V rail for the ESP32 and sensor front-end and a separate high-current rail for the L293D motor drivers, isolating the noise-sensitive digital logic lines from the inductive voltage ripple and stall-current spikes of the brushed DC actuators during peak load conditions [48], [21].

To extend operational battery life during periods of prosthetic inactivity, the ESP32's native Deep Sleep mode was implemented in firmware. In this state, all peripheral clocks, the Wi-Fi radio, and the main CPU cores are gated off, reducing the microcontroller's current draw from approximately 240 mA during active multi-sensor polling to under 10 µA. A configurable inactivity timer monitors the sEMG envelope and armband pressure channels; if no voluntary activation event is detected within a user-defined timeout window, the system autonomously transitions to Deep Sleep and resumes full operation upon the next muscle contraction trigger via the ULP co-processor wake-up interrupt. This adaptive duty-cycling strategy significantly reduces mean current consumption during daily prosthetic use, where active grasping intervals are interspersed with extended periods of limb rest [48].

Fig. 18.

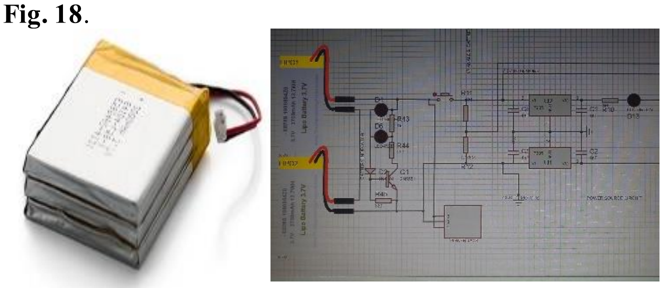

Fig. 18. Schematic of the power management unit showing the 7.4V LiPo source and dual-stage buck regulation paths for digital and motor power rails.

4) Software Architecture and Multi-Modal Control Topology:

The ESP32 firmware implements a layered multi-modal control pipeline in which four operating modes share a common actuation scheduler [46], [50]:

Mode 1 — Pure EMG Control: evaluates the rectified, smoothed sEMG amplitude directly and maps temporal intensity thresholds to discrete motor actions, providing a fallback channel when the other sensors are unavailable.

Mode 2 — Proximity-Triggered IR and Force Regulation: uses the IR proximity channel to pre-shape the optimal grip profile before physical contact occurs, then hands over to the four fingertip FSRs (6 mm × 6 mm each) for compliant force modulation during contact [27], [36].

Mode 3 — Kinematic IMU Orientation Mapping: translates residual-limb tilt into grip configuration through gravity-compensated positioning — forward tilt drives uniform finger flexion, backward tilt drives extension, and a dead-band around level orientation locks the actuators — providing orientation awareness independent of EMG quality [31], [32].

Mode 4 — Asynchronous Sensor Fusion (Primary Mode): combines all channels through a deterministic gating logic in which the moving-average-filtered EMG envelope acts as an asynchronous activation trigger, the Velostat armband pressure matrix disambiguates grasp intent, the MPU6050 contributes kinematic shape selection, and the fingertip FSRs close the loop with a compliant termination command once a stable grip is established [29], [27], [36], [24]. This layered fusion architecture ensures modularity and facilitates rapid integration of emerging sensing modalities [51], [52], [53].

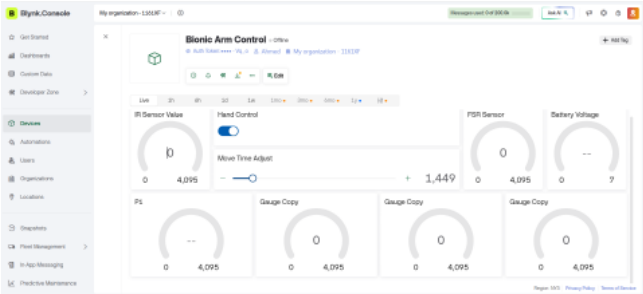

Fig. 19. Software ecosystem architecture illustrating the real-time data flow between the ESP32, Bolt Cloud library, and cloud web portal.

Fig. 20. Mobile application interface showing grip mode selection, real-time EMG threshold display, and sensor status panel.

III. RESULTS AND DISCUSSION

Longitudinal Prototype Evolution Case Study (2022–2026)
To achieve the optimized biomechatronic profile of the AB-Limb, a four-year longitudinal development case study was conducted across three distinct generational prototypes, benchmarking structural durability, weight reduction, and sensor reliability at each stage.

Prototype I (Conceptual Phase — 2022): Focused on basic multi-finger kinematics using a fully 3D-printed PLA frame with analog IR sensors. Testing revealed critical limitations in structural strength under load, high inertial mass, and significant electrical noise that precluded reliable EMG signal acquisition.

Prototype II (Sensory Integration Phase — 2024): Transitioned structural framing to reinforced polymers and aluminum alloys. Custom active surface EMG bio-amplifiers and commercial FSRs were introduced. Grip force control was successfully validated; however, structural mass remained high and raw EMG drift produced false-positive actuator triggering, motivating the dual-verification approach adopted in the final design.

Prototype III (Optimized Final Platform — 2026): The structural framework was fully transitioned to VARTM-molded carbon fiber-epoxy composite, integrating the 2×2 Velostat armband scanning matrix, MPU6050 orientation tracking, and autonomous reflex gating logic, as shown in Fig. 21. This iteration achieved a 40% reduction in structural weight relative to Prototype II and a production cost below \$310 USD, representing over 90% reduction compared to premium commercial benchmarks.

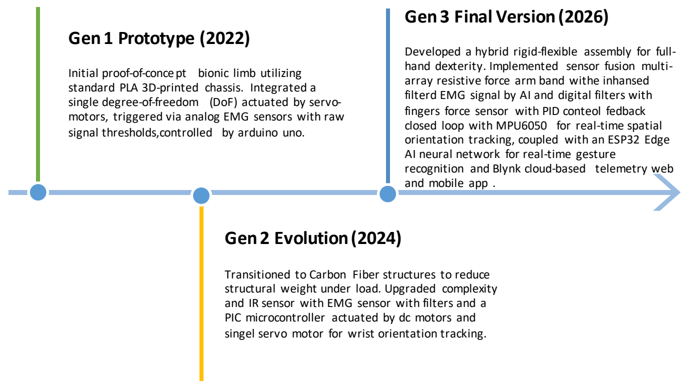

Fig. 21. Generational evolution chart of the AB-Limb prototypes (2022–2026) mapping structural material transitions against sensor integration complexity.

B. System Actuation and Latency Performance

End-to-end system latency — defined as the elapsed time from initial voluntary muscle contraction to completion of terminal device grasp execution — was assessed using repeated timed trials with a mobile stopwatch reference across multiple contraction-and-release cycles. The measured average deployment latency was 380 ms under normal operating conditions, which sits comfortably below the 500 ms perceptual threshold identified in the literature as the boundary for intuitive prosthetic responsiveness [54], [56].
The PID velocity control loop was evaluated by comparing motor transient behavior with and without active PID feedback during repetitive finger flexion cycles. Without PID correction, the N20 motors exhibited visible positional oscillation and overshoot near object boundaries. With PID active, convergence to the target position was smooth and asymptotic, with no observable overshoot, as illustrated in Fig. 22. The PID response curve was generated from firmware timestamp logs processed in MATLAB to reconstruct the motor velocity profile over time [57].

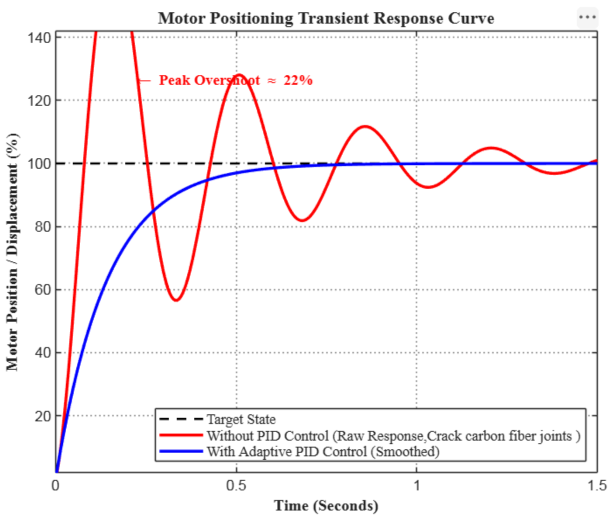

Fig. 22. Simulated transient response curve illustrating motor positioning behavior with and without PID feedback, generated from firmware timing parameters in MATLAB.

C. Multi-Modal Sensor Validation and Feedback Stability:

The custom LM324-based instrumentation amplifier front-end was evaluated using an oscilloscope during dynamic arm movement. The baseline noise floor was observed to fluctuate during measurement; a representative peak-to-peak noise level of approximately 15 mV was recorded under typical operating conditions, consistent with acceptable surface EMG acquisition noise thresholds reported in the literature [1], [2]. The dual-verification gating logic — requiring simultaneous EMG threshold crossing and armband confirmation — effectively suppressed false-positive actuator triggers during testing.

The 2×2 Velostat armband pulse-scanning method successfully eliminated ghosting artifacts during multi-point simultaneous contact. The custom 6 mm × 6 mm FSR sensors were evaluated by applying manually-varied compressive loads ranging from light fingertip pressure to a 4 kg static weight. The sensors produced a monotonically decreasing resistance-to-load response, and the Force-to-Velocity mapping algorithm successfully modulated motor PWM proportionally — a filled plastic cup was gripped without deformation, while a paper cup was grasped without crushing, demonstrating adaptive compliance under functional load variation. The PWM-driven vibration motor activated reliably upon fingertip FSR contact during all trials; precise latency was not instrumented in this prototype iteration and is reserved for future quantitative evaluation.

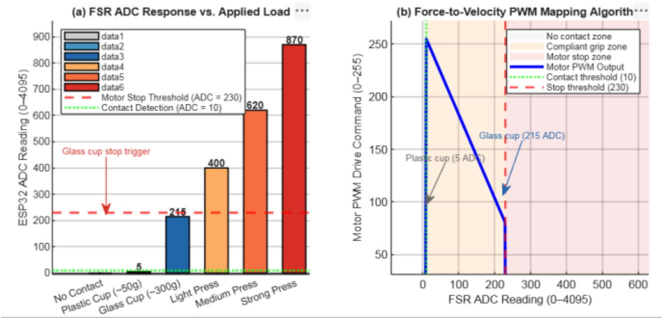

Fig. 23a. Functional grasping validation: the AB-Limb successfully gripping a filled plastic cup (left) and a paper cup (right) without deformation, demonstrating adaptive Force-to-Velocity compliance.

Fig. 23b. FSR characterization and Force-to-Velocity algorithm validation: (a) measured ESP32 ADC responses across representative applied loads — a plastic cup produced 3–7 ADC units, a glass cup produced 200–230 ADC units, and direct finger compression yielded 400–900 ADC units; (b) the corresponding Force-to-Velocity PWM mapping showing progressive motor deceleration from full drive to complete stop as grip force increases, with the motor halt threshold set at ADC = 230 to prevent object deformation.

D. Socioeconomic Cost Analysis and Weight Optimization:

A core design objective of the AB-Limb is substantial cost reduction without compromising biomechatronic functionality. All component purchases were tracked through itemized invoices totaling approximately 15,000 EGP (~$310 USD at current exchange rates), providing an auditable cost basis for the prototype. Advanced commercial myoelectric hands such as the BeBionic (Otto Bock) and I-Limb (Össur) are priced at approximately €28,000, while the Vincent Hand reaches $44,500, placing them beyond the economic reach of the majority of prosthetic users in low- and middle-income countries [4], [58]. In line with WHO prosthetics and orthotics provision guidelines [5], the AB-Limb targets this critical affordability gap.

Table I presents a comprehensive technical and economic benchmark comparing the AB-Limb against three premium commercial alternatives and the open-source compliant hand by Akhtar et al. [10].

TABLE I
Socioeconomic and Structural Comparison of Upper-Limb Prosthetics

Technical Specifications	Be Bionic [4]	I-Limb [4], [8]	Vincent Hand [4]	Open-Source Compliant Hand [10]	Proposed AB-Limb
Manufacturing Company	Otto Bock	ÖSSUR	Vincent Systems	A. Akhtar et al. (UIUC)	A.B. Labs
Approximate Cost	€28,000	€28,000	$44,500	$553.06 USD	~$310 USD (15,000 EGP)
Total System Weight (g)	500g	599g	410g	~425g (Hand Only)	350g
Operating Voltage (V)	7.4V	7.4V	6.0V – 8.0V	7.4V	7.4V
Battery Chemistry	Li-ion	Li-Po	Li-Po	Li-Po	Li-Po
Battery Capacity (mAh)	1300 – 2200	1300 – 2400	1300 – 2600	1500	1500 – 3000
Total Actuator Count	5	6	6	5	7
Actuator Drive Type	DC Motor + Lead Screw	DC Motor + Worm Gear	DC Motor + Worm Gear	Mini DC Worm Gear	Mini DC Worm Gear & Stepper
Active Digit Count	5	5	5	5	5
Grip Force Range (N)	140 N	100 N	60 N	–	51 N – 132 N
Control Modalities	EMG	EMG, Gesture, Mobile App	EMG	Commercial EMG, Force	EMG, IR, IMU, Force Matrix, App, Web

As shown in Table I, the AB-Limb achieves a disruptive price point of ~$310 USD, representing a 90% cost reduction relative to commercial alternatives [4]. At 350 g, it is the lightest design in the comparison, with seven independent actuators exceeding all compared systems. Control versatility spans six input modalities (EMG, IR, IMU, Force Matrix, App, and Web), compared to the single or dual modalities of commercial alternatives.

E. Layered Control Architecture and Smart Gating Validation:

The embedded control hierarchy was validated across three functional real-time operational layers through structured physical testing, as illustrated in Fig. 24.

1) Reflexive Layer: The palm-mounted IR proximity sensor was calibrated via a trimmable potentiometer integrated into the circuit, set to trigger at approximately 3 cm object distance to suppress spurious background reflections. Functional validation confirmed that placing the hand near an object during a hold state reliably triggered an emergency close command through the hardware interrupt path, bypassing the main EMG processing thread and mimicking a biological protective reflex response.

2) Precision Layer: Fingertip FSR force feedback was validated using two representative test objects: a filled plastic cup and an empty paper cup. The Force-to-Velocity algorithm successfully modulated motor drive current in both cases — maintaining a stable grip on the filled cup without slippage and preventing structural deformation of the paper cup. These results confirm that the closed-loop proportional force feedback layer provides effective adaptive compliance across a functionally relevant load range [60], [59].

3) Stabilization Layer: The IMU-based leveling mode was validated in a structured cup-retention task. The prosthetic hand was commanded to grip a water cup placed on a flat surface with finger actuators locked in the closed position. The forearm assembly was then tilted laterally left and right — the wrist stepper motor compensated in the opposite direction to maintain the cup's vertical orientation. The forearm was subsequently tilted forward, causing the fingers to open and lower toward the cup surface; upon lowering the wrist, the fingers closed and successfully re-grasped the cup. This trial confirmed that the MPU6050-driven gravity-compensated positioning delivers functional object stabilization independent of EMG input [5], [48], [49].

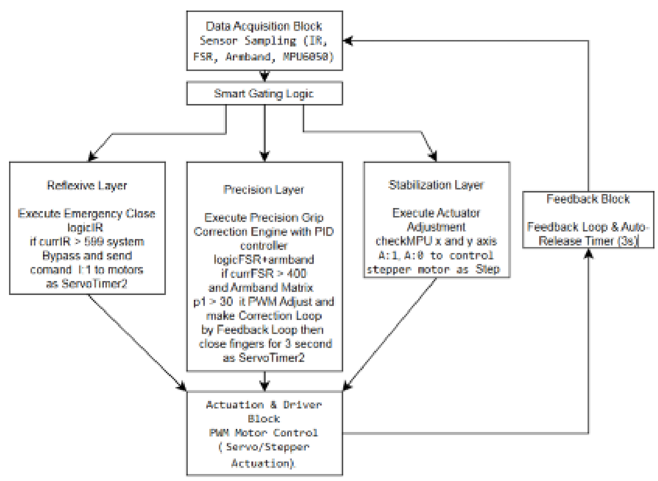

Fig. 24. System control flowchart illustrating the interaction between the Reflexive, Precision, and Stabilization layers.

IV. CONCLUSION

This paper presented the design, structural optimization, and multi-modal control architecture of the AB-Limb, a low-cost, high-functionality upper-limb bionic prosthesis. By leveraging a custom-fabricated carbon fiber-epoxy composite framework via VARTM, the complete forearm and hand assembly achieved a structural mass of 350 g — a 40% reduction relative to conventional aluminum or 3D-printed PLA equivalents. A deterministic hybrid control pipeline on the ESP32 microcontroller eliminated classification latencies, achieving a measured end-to-end deployment response time of 380 ms, well within the 500 ms intuitive control threshold.

Economically, the substitution of premium commercial modules with custom-engineered EMG electronics, miniaturized Velostat sensor configurations, and open-source mechanical components constrained the total production cost to approximately $310 USD (15,000 EGP), validated by itemized purchase invoices. Future work will focus on instrumented quantification of vibration feedback latency, formal clinical evaluation with prosthetic users, integration of continuous glucose monitoring sensors within the socket interface, and deployment of edge-AI regression algorithms for proportional myoelectric velocity control directly on the ESP32.

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